

TANISH PAL

Electronics Engineer — Digital Design — ASIC Design Flow — C/C++ — Python

📞 +91-7009890319 | ✉ tanish.pal.biz@gmail.com | 🔗 LinkedIn |

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

B.Tech in Electronics and Communication Engineering

2022 – 2026

CGPA: 7.21/10

TECHNICAL SKILLS

Digital Design: Digital Logic Design, Combinational & Sequential Logic, FSMs, Registers, Counters, Computer Architecture

ASIC Fundamentals: ASIC Design Flow, RTL Design Concepts, Logic Synthesis Fundamentals, Design Constraints, Timing Analysis Fundamentals

Programming: C, C++, Python, Object-Oriented Programming, Data Structures & Algorithms

Scripting: Python, Unix/Linux Shell, Tcl Fundamentals

Hardware: Microcontrollers, Embedded Systems, STM32, UART, GPIO, Interrupts, Firmware Debugging

Tools: Git, GitHub, STM32 Development Tools, Linux/Unix Environment

RELEVANT COURSEWORK

Digital Electronics, Digital Logic Design, VLSI Design, Computer Architecture, Microprocessors and Microcontrollers, Embedded Systems, Semiconductor Devices

PROJECTS

UART-Based Device Control System using STM32 | *Embedded C, STM32, UART, GPIO, Interrupts* May 2025 – Jun 2025

- Developed Embedded C firmware for UART-based communication between a host PC and an STM32 microcontroller.
- Implemented command-driven GPIO control and device status reporting through a serial communication interface.
- Configured UART and GPIO peripherals and implemented interrupt-based communication for responsive command processing.
- Debugged firmware using serial-terminal testing and validated communication, command handling, peripheral control, and reset behavior.

RTL Design and Synthesis of Digital Logic Blocks | *Verilog, RTL, Synthesis, Tcl*

- Designed and verified synthesizable RTL modules including combinational logic, sequential logic, multiplexers, registers, counters, and finite-state machines using Verilog.
- Developed simulation testbenches to validate RTL functionality across normal, boundary, and reset conditions using waveform-based debugging.
- Synthesized RTL designs and analyzed generated reports for area, timing, cell utilization, and critical paths to understand implementation trade-offs.
- Used Tcl-based scripts to automate repetitive design-flow tasks including compilation, simulation/synthesis execution, report generation, and result collection.
- Studied RTL-to-gate-level transformation and the impact of design constraints on synthesis results and timing behavior.

RTL Equivalence and ECO Analysis | *Verilog, Python, Digital Logic*

- Implemented functionally equivalent versions of RTL logic and analyzed output behavior across different RTL implementations.
- Compared pre- and post-modification designs using simulation-based checks to identify functional mismatches introduced by design changes.
- Studied common ECO scenarios and analyzed how localized RTL changes can affect synthesized logic, timing, and functional behavior.
- Developed Python utilities for processing simulation outputs and organizing design verification results.

EXPERIENCE

A2IT / Cheffer

Jun 2025 – Jul 2025

Software Development Intern

- Developed reusable software components and integrated REST APIs while debugging application-level issues.
- Worked collaboratively on implementation and optimization of application components.

ACHIEVEMENTS

Solved 500+ problems on LeetCode with a peak contest rating of 1726.